MILITARY SPECIFICATION

TELESCOPES, ELBOW M92D AND M92F:

1. SCOPE

1.1 This specification covers Elbow Telescopes. M92D and M92F used for direct firing with the 106 MM Recoilless Rifle.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-O-13830 —Optical Components of Fire Control lnstruments: General Specification Governing the Manufacture and Inspection of

MIL-F-13926 —Fire Control Materiel: General Specification Governing the Manufacture and Inspection of

MIL-P-14232 —Parts, Equipment and Tools for Ordnance Materiel, Packaging of

MIL-I-45208 —Inspection Requirements, General Specification for

MIL-I-46607 —Inspection Equipment, Supply and Maintcnance for Ordnance

STANDARDS

MILITARY

MIL-STD-105 —Sampling Procedures and Tables for inspection by Attributes

MIL-STD-130 —Identification Marking of U. S. Military Property

DRAWINGS

MUNITIONS COMMAND

F8299680 —Telescopes, Elbow: M92D and M92F

IEL-8286960 —Index of Inspection Equipment .List for Telescope, Elbow: M92F

IEL-8299680 —Index of Inspection Equipment List for Telescope, Elbow: M92D

LSQAP-8286960 —Supplementary Quality Assurance Provisions List for Telescope, ELbow : M92F

L~SQAP-8299680 —Supplementary Quality Assurance Provisions List for Telescope, Elbow: M92D

(Copies of specifications, standards, drawings and publications required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 initial production sample. Unless otherwise directed by the contracting officer, an initial productions sample shall be required. The sample shall be manufacture in the same manner, using the same materialist equipment, processes and procedures as will be used in regular production.

3.2 Materials. Materials shall be in accordance with drawings, material specifications

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and general specifications forming a part of this specification.

3.3 Fabrication. The telescope shall be manufactured in accordance with the drawings applicable to the type telescope specified in the contract or order and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired optical and mechanical characteristics, performance and permanence to insure inherent quality and performance capabilities.

3.4 General specifications. The contractor shall be responsible for adherence to and compliance with the following requirements of Specifications MIL-0-13830 and MIL F-13926.

3.4.1 *MIL-0-13830.*
(a) Cleanliness
(b) Illuminated reticle

3.4.2 *MIL-F-13926.*
(a) Order of precedence
(b) Protective finished surfaces
(c) Special accuracy
(d) Finish, painting and protective
(e) Workmanship

3.5 Environmental.

3.5.1 *Storage tcmpcratures.* The telescope shall show no evidence of physical failure and shall meet all requirement of this specification at standard ambient temperature (pros 60” to 90° Fahrenheit) after having been exposed and thermally stabilized at ambient temperatures of plus 160° and minus 80° Fahrenheit.

3.5.2 *Vibration*. The telescope shall be vibrated in a vertical plane at a constant frequency of 30 cycles per second with an amplitude of 1/10 inch (1/8 inch maximum excursion) for a period of 5 minutes plus or minus 15 seconds. Subsequent to vibration the telescope shall show no evidence of loose or damaged parts and shall meet the requirements of 3.6 to 3.10 inclusive.

3.6 Cleanliness . The telescope shall meet the cleanliness requirements of Specification MIL-0-13830. The surfaces of the reticle shall show no more than two particles of dirt. The area of each particle shall not be in excess of an area equivalent to a size two dig. There shall be no other dirt within the telescope which shall adversely affect observation and image quality.

3.7 Sealing. The telescope shall show no evidence of leakage when subjected to an external pressure of dry nitrogen at five pounds per square inch gage pressure (PSIG) for a period of at least five minutes, Dewpoint of nitrogen shall be at least as low as minus 25° Fahrenheit.

3.8 Accuracy.

3.8.1 *Collimation.* The line of sight through the center of the boresight cross shall be 20 mils plus or minus 0.5 mil above the geometric axis and shall coincide within 0.1 mil of a plumb line passing through the geometric axis of the telescope.

3.8.2 *Parallism of image and reticle.* With a line of sight parallel to a horizontal plane formed by the 90 degree geometric axis of the telescope body, the plane through the center lines of the keyways vertical, and the cross level vial centered, the horizontal reticle line shall not deviate from the image of the horizontal target reference line by more than the anglc subtended by 0.4 mil at one extremity of the horizontal reticle line, when the other extremity of the line is in coincidence with the reference line.

3.8.3 *Eyepiece focus.* The eyepiece focus shall be fixed between the limits of minus 0.75 and minus 1.0 diopter.

3.8.4 *Resolution.* The resolution on the optical axis shall be fifteen seconds of arc or less.

3.8.5 *Parallax.* Parallax at the geometric axis shall not exceed 0.10 mil when viewing a target at 125 yards plus or minus 20 yards.

3.8.6 *Reticle accuracy.* The angular value for the reticle graduation between zero and maximum range shall read as indicated on the reticle drawing within plus or minus 0.25 mil.

3.9 Illumination. The reticle markings shall

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appear clearly defined when observed in a darkened area, utilizing Instrument Light M42 or equivalent.

3.10 Identification marking. Identification marking, unless otherwise specified shall be in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibilities. The supplier is responsible for the performance of all inspection requirements as specified herein, Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General requirements. Inspection requirements and procedures shall be in accordance with Specification MIL-I-45208.

4.3 Initial production sample. Prior to production, the contractor shall submit the initial production sample in accordance with 3.1 for inspection by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The sample shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

4.3.1 *Initial production sample failure.* Should the production sample fail to pass the prescribed tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. When corrections in the process have been made, a new production sample shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production sample is submitted and corrective actions have been taken to eliminate deficiencies.

4.3.2 *Control samples.* After acceptance of the initial production sample, one telescope from every 100 completed during production, or one each month, whichever occurs first, shall be examined for compliance with 3.2 and 3.3.

4.4 *Lot formation.* Inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.4.1 *Defective units and lots.* Defective units. and lots shall be processed as specified in Standard MIL-STD-105.

4.5 Material. Control of material shrill be in accordance with test data requirements of Specification MIL-I-45208.

4.6 Inspection of components. Inspection requirements for component parts and subassemblies are contained herein and on Supplementary Quality Assurance Provisions (SQAP) listed in Supplementary Quality Assurance Provisions List (LSQAP)-8286960 for Telescope Elbow M92F or (LSQAP)-8299680 for Telescope Elbow M92D.

4.6.1 *Contents of SQAP.* The contents of SQAP shall represent the minimum inspection requirements and shall be the basis for acceptance or rejection by lots of each component covered by a SQAP. The Government reserves the right, however, to inspect for the requirements of all drawings and specifications and to reject individual nonconforming items.

4.6.2 *Contractor's inspection for all components not covered by SQAP.* Inspection for these components shall be in accordance with the inspection procedures of the inspection system required by Specification MIL-I-45208. The system shall include procedures for checking all positional and functional characteristics shown on the drawings which indicate precision requiremnts and have a bearing on the functional accuracy and performance of the telescope.

4.6.3 *Inspection equipment.* Inspection equipment shall conform to drawings listed in IEL-828690 for Telescope M92F and

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IEL-8299680 for Telescope M92D except as provided for under the alternative inspection procedures and inspection equipment requirement of Specification MIL-I-45208. Provisions of Specification MIL-I-45607 shall apply to test equipment provided by the Government or provided by the contractor and utilized in the Government approved inspection system, The Government reserves the right to use all approved test equipment both Government furnished and contractor acquired, for verification inspection.

4.6.4 *Accuracy requirements.* Inspection equipment which incorporates features for measuring values or sizes established by product requirements shall be capable of measuring those values within an accuracy of ten percent of the specified tolerance, except as otherwise specified. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.7 Acceptance inspection.

4.7.1 *Classification of defects.* The classification of defects in tables I and II shall constitute the minimum inspection requirements to be performed prior to acceptance or rejection by lots, The AQL specified pertains to each individual characteristic Samples selected for acceptance inspection listed under table II shall include those samples used for acceptance inspection specified in table I.

4.7.1.1 *Disposition of rejected lots.* Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lots shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

TABLE I *Classification of defects.*

USE inspection level II of table IIIA with sampling plan table IV-A of Standard MIL-STD-105.

Critical: None defined.

Major: AQL 1.0 percent defective	Requirement paragraph	Q.A. paragraph	Test method
101. Collimation	3.8.1	4.8.2.1	Special Testing Equipment (STE)
102. Parallelism of image and reticle	3.8.2	4.8.2.2	STE
103. Eyepiece focus	3.8.3	4.8.2.3	Standard Meas. Equipment (SME)
104. Resolution	3.8.4	4.8.2.4	STE-Visual
105. Parallax	3.8.5	4.8.2.5	Visual
106. Reticle accuracy	3.8.6	4.8.2.6	SME
107. Illumination	3.9	4.8.3	Visual
108. Identification marking	3.10	4.8.4	Visual
Minor: None defined.			

Note. The tests of table I shall be performed at standard ambient temperature (plus 60° to 90° Fahrenheit).

TABLE II. *Classification of defects*

Use inspection level L-6 of table IIIB with sampling plan table IVA of Standard MIL-STD-105.

Critical: None defined.

Major: AQL 4.0 percent defective	Requirement paragraph	Q.A. paragraph	Test method
101. Storage temperatures	3.5.1	4.8.1.1	Special Testing Equipment (STE)
102. Vibration	3.5.2	4.8.1.2	STE
103. Cleanliness	3.6	4.8.1.3	Specification MIL-O-13830
104. Sealing	3.7	4.8.1.4	Standard Meas. Equipment (SME)
Minor: None defined.			

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4.8 Test methods and procedures.

4.8.1 *Environmental.*

4.8.1.1 *Storage temperatures.* This test shall be performed in accordance with procedure II of Specification MIL-F-13926 at the specified temperatures. When the telescope is thermally stabilized at room temperature it shall be. tested for compliance with 3.5.1.

4.8.1.2 *Vibration.* The telescope shall be mounted on a vibration fixture equal to or better than vibration fixture F7560085. The vibration shall be at the amplitude, frequency and duration specified. Subsequent to the test, the telescope shall meet the requirements of 8.5.2.

4.8.1.3 *Cleanliness.* The optical surfaces shall be inspected for cleanliness in accordance with Specification MIL-O-13830. The size and quantity of dirt particles shall not exceed the requirements of 3.6

4.8.1.4 *Sealing.* The telescope shall be pressure tested in accordance with the setup and operating instructions on Drawing F7680673 and shall meet the requirements of 3.7.

4.1.8.2 *Accuracy.*

4.8.2.1 *Collimation.* Collimation test shall be accomplished by means of a final inspection and collimation fixture depicted on Drawing F7681055 and listed on IEL 828-6960 and IEL-8299680. The telescope shall be positioned in a horizontal plane with the Keyway on top. When the telescope is properly oriented, the angular subtense of the angle formed by projecting the line of sight through the boresight cross and a line of sight coincident to the intersecting point of the target plumb line and the horizontal crossline shall not exceed the value specified in 3.8.1 as measured on the target.

4.8.2.2 *Parallelism of image and reticle.* The telescope shall be positioned as specified above and the cross level vial centered; one extremity of the horizontal reticle line shall be coincident to the target line, The angle subtended by the other. extremity of the horizontal reticle line shall be parallel to the horizontal geometric axis within the limits specified in 3.8.2 as measured on the target.

4.8.2.3 *Eyepiece few.* Eyepiece focus test is made with the aid of a 3-power dioptometer adjusted for the observer's. eye accommodation. Locate the telescope with the objective facing an illuminated white background so that the reticle pattern is clearly visible. Place the dioptometer in front of the eyepiece and remove parallax between reticles, by movement of the dioptometer scale. The reading of the dioptometer scale will indicate the error which shall be within the limits of 3.8.3.

4.8.2.4 *Resolution.* Resolution test is accomplished with the same setup used in 4.8.2.1. The test shall be in accordance with Specification MIL-O-13830 and shall meet the requirements of 3.8.4.

4.8.2.5 *Parallax.* This test shall be accomplished by sighting the telescope on a target at the distance specified in 3.8.5. Paral lax is recognized as any apparent movement of the recticle lines in relation to the target's image when the observer's head is moved side to side or up and down. Parallax at the geometric axis shall not exceed the tolerance specified in 3.8.5.

4.8.2.6 *Reticle accuracy.* Reticle accuracy test for the angular value of the graduation marks is accomplished by means of a mil block chart with dimensions suitable for the target distance utilized and based on the dimensional notes given on Drawing D821-3863. The error for this test shall not exceed the amount indicated in 3.8.6.

4.8.3 *Illumination.* Illumination test is a visual check made in a darkened area as indicated in 3.9.

4.8.4 *Idcntification marking.* Identification marking test is visual and made in accordance with Standard MIL-STD-130.

4.9 Packaging, packing and marking. Quality Assurance provisions for packaging, packing and marking shall be in accordance with Specification MIL-P-M1432.

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5. PREPARATION FOR DELIVERY

5.1 Packaging packing and marking. The telescope shall be cleaned, preserved, packaged, packed and marked in accordance with Specification MIL-P-14232/P8299680. The level of protection shall be as specified in the procurement document.

6. NOTES.

6.1 Intended use. These elbow telescopes are intended as direct fire telescopes for the 106 mm Recoilless Rifle. The Telescope M92D has a reticle inscribed for ranges in yards. The telescope M92F has a reticle inscribed for ranges in meters. These telescopes are mounted in Telescope Mount M90.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of an applicable level of preservation, packaging and packing.
(c) Applicable stock numbers.
(d) Provisions for initial production sample including date of delivery (see 3.1).

6.3 Design information. Information regarding specific design not covered or referenced by this specification may be obtained from the procuring activity upon specific request of the contractor.

6.4 Definitions.

6.4.1 *Standard measuring equipment (SME).* Standard measuring equipment (SME) is defined as the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are nornmally used by an inspector to perform dimensional inspection of items under procurement, This category also includes commercial testing equipment such as meters, optical comparators, etc.

6.4.2 *Special testing* equipment (STE), Special testing equipment is defined as special inspection fixtures, instruments, and modified standard measuring equipment.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in an manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any invention that may in any way be related thereto.

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